

Does the rat body predict human harm?

~90% of drugs that clear animal safety still fail in humans, a third of it toxicity. We asked whether **rat liver gene expression** adds anything to **chemical structure** when predicting toxicity — and stress-tested every promising signal until it broke or held.

BOTTOM LINE

Gene expression helps a little, but the wall is **cross-dataset comparability**, not data volume — and for real human liver injury, structure alone still wins.

↗ [GitHub](#) — code & pipeline

↗ [HuggingFace](#) — data & results

TOX21

Does rat expression add to structure?

Yes — small, stress-response-specific

SCALING

Does more animal data help?

No — comparability is the ceiling

HUMAN DILI

Predict real clinical liver injury?

No — structure alone wins

WITHDRAWAL

Predict drugs pulled from market?

Biology edges it — but noisy

01

The hypothesis

Toxicity is where the animal→human translation gap bites hardest. Chemical structure already predicts many toxic mechanisms — but structure is blind to *what a dose actually does to a living liver*. Gene expression is that readout.

Core question. Given a compound's SMILES, does adding the rat liver's transcriptional response improve prediction of the 12 Tox21 human toxicity-mechanism assays?

Sharper sub-hypothesis. Expression should help the **stress-response** mechanisms — which act *through gene regulation* and leave a transcriptional footprint — more than the **nuclear-receptor binding** mechanisms, where the molecule's shape already *is* the answer. Call it the **SR > NR** prediction.

02

The data

Two rat toxicogenomics resources, cross-referenced to a human assay panel by standardized chemistry.

672

unique rat compounds
TG-GATEs ∪ DrugMatrix

613

carry a Tox21 label
the full labelled cohort

177

primary supervised set
clean single-platform liver expression

256

pooled with TG-GATEs
177 + 79 new · ComBat-merged

Reading the ladder. 613 compounds have Tox21 labels, but clean expression exists for only **177** — the Affymetrix Rat 230–2 liver arm, chosen so DrugMatrix and TG-GATEs share a platform (coverage traded for a clean batch correction). Pooling a second rat source was *expected* to reach ~271, but the single-dose fetch realised **256** — only 79

genuinely new compounds. Matching throughout is on InChIKey connectivity (desalt → neutralise → key), identical across all three sources.

NUCLEAR-RECEPTOR · 7

Receptor binding — a function of molecular shape.

NR-AR

NR-AR-LBD

NR-AhR

NR-Aromatase

NR-ER

NR-ER-LBD

NR-PPAR γ

STRESS-RESPONSE · 5

Cellular damage programs — expressed through genes.

SR-ARE

SR-ATAD5

SR-HSE

SR-MMP

SR-p53

03

The pipeline

Modular, config-driven, and leakage-safe end to end — each stage independently runnable and cached.

curate	signature	correct	reduce	fuse	evaluate
Link & key InChIKey match across TG-GATEs, DrugMatrix, Tox21	→ logFC treated – matched vehicle, high-dose liver	→ ComBat batch = data source; validated on shared drugs	→ PCA · 100 fit on the train fold only	→ Structure ⊕ GE z-scored blocks, balanced	→ Model per + A

Every data-dependent transform – PCA, scalers, ComBat, the frozen Tox21 head – is **locked** fit inside the training fold only. Repeated stratified cross-validation; fixed seeds; splits persisted.

HOW THE MODELS ACTUALLY WORK

The plan named a "masked multitask MLP head." What actually ran is simpler, more honest at this sample size, and fully interpretable — **no neural net was trained.**

```
// features → prediction (all transforms fit in the train fold only)
SMILES      → ECFP4 · 2048-bit → PCA·128 →
```

logFC · 31k probes → PCA · 100 ————— | z-score → concat → Logistic × 12
(one per assay, masked)

01 Not one model — twelve

One independent classifier *per Tox21 assay*, each trained only on the compounds where that assay was measured. Tox21's sparse labels are handled by *subsetting*, not a masked loss — same guarantee, no network needed.

The cost: no parameter sharing. A sparse assay (NR-PPAR γ , 12 actives) can't borrow strength from a dense one (SR-ARE, 99) — which is the *entire* theoretical case for the MLP.

02 A linear head, on purpose

LogisticRegression · L2 · class-balanced . At $N \approx 140$ /fold with ~228 features, the regularised linear model is the stable choice, and balancing is what makes AUC — not accuracy — the honest metric on a 5–27%-active problem.

It also keeps the science interpretable: "does expression add?" is just "do the GE components earn nonzero weight beside structure?"

03 Why not the fancier models

Gradient-boosted trees (Baseline 2) were the high-capacity test — and they *overfit*: $0.733 < 0.757$. Flexibility is a liability when data is scarce, which is the empirical case against a neural net too.

The joint multitask MLP is **deferred, not rejected** — blocked by the Intel-Mac torch stack, documented as a swap-in for a torch-capable machine.

04 No leakage, anywhere

Both scalars, both PCAs, ComBat, *and* the frozen Tox21 model feeding Validation B's Approach A are fit inside the training fold only.

That discipline is exactly what made the Approach-A leakage *visible* — cross-fitting caught it and collapsed a fake 0.646 to a real 0.465.

Experiments & results

Each finding is paired with the check that could have killed it. Verdicts are color-coded:

signal

nuanced

null.

PART 1 – DOES GENE EXPRESSION ADD? (TOX21, N=177)

R1

SIGNAL

Rat expression adds a small, stress-response-specific benefit.

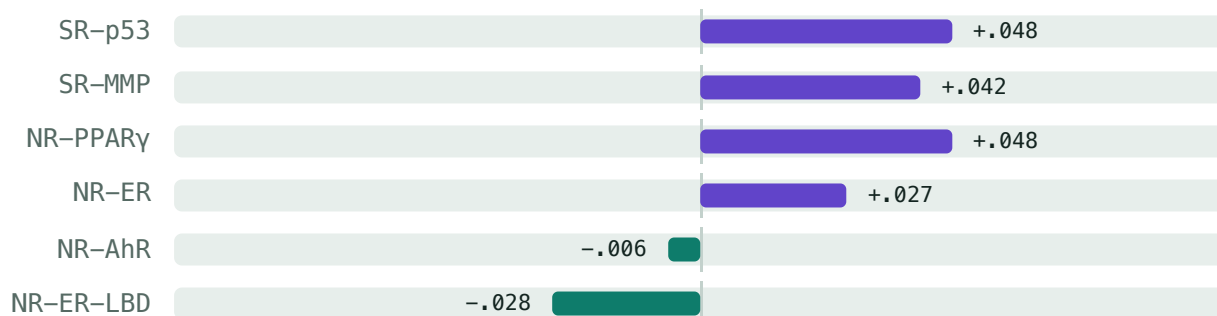
Three models, identical except the feature block. Gene expression *alone* already beats chance — proof the rat transcriptome carries human-toxicity signal.

0.757

structure only (AUC)

0.679

expression only

0.766structure +
expression**+0.026**SR-panel Δ AUC

Why. Expression helps stress-response and the two *transcriptionally active* receptors (PPAR γ , ER are ligand-activated transcription factors), and does nothing for the pure ligand-binding assays (AR-LBD, ER-LBD) where structure already hits ~0.90. The SR > NR prediction holds.

R2

GOTCHA CAUGHT

The benefit only appears once the two feature blocks are balanced.

Before z-scoring the PCA blocks, structure's high-variance components drown out the 100 expression dimensions and fusion loses. Documented as a before/after so the win can't be dismissed.

-0.041

macro Δ AUC, blocks
unbalanced

+0.009

macro Δ AUC, z-
scored

Why. An L2 head penalizes all coefficients equally, so an unscaled block with larger variance is effectively ignored. This is exactly the "expression drowned out" artifact — and why the effect is real but fragile.

R3 · R4

HOLDS UNDER STATS

100 components is optimal, and SR > NR is significant at the repeat level.

The stress-response benefit grows monotonically with the number of expression components — it's distributed signal, not overfit noise. And the panel contrast survives a formal test.

+0.026

SR – NR panel Δ AUC

[.014, .037]

95% CI — excludes 0

0.074

assay-level Mann-
Whitney p

Why. The repeat-level test is significant; the assay-level test is a strong trend (only 12 assays, and PPAR γ /ER break the clean split for a good biological reason). The real divide is *"acts through gene regulation" vs "acts through binding."*

The benefit is model- and encoder-dependent — and ECFP beat ChemBERT.

Two robustness probes. Gradient-boosted trees overfit at N=177 and erase the gain. A frozen ChemBERT structure encoder turned out *weaker* than plain ECFP fingerprints.

0.733

GBM structure
(overfits) vs 0.757

0.681

ChemBERT structure
vs 0.757 ECFP

robust

SR > NR holds under
both encoders

Why ChemBERT loses — a coordinate mismatch, not a knowledge gap. ECFP is a sparse, axis-aligned basis where each bit *is* a substructure, so a linear head reads "has reactive group X → active" straight off — exactly how these assays fire. A frozen transformer stores the same facts in a *dense basis rotated for a different objective* (reconstructing SMILES tokens); re-isolating "has a nitroaromatic" from 768 entangled dimensions is a learning problem you can't solve at N=177. So the loss is *expected* — and it **strengthens** the study: "does GE add?" ran against a genuinely strong, literature-appropriate baseline, not a weak one. It fires especially cleanly here because SR toxicity is driven by *reactive structural alerts* — the very substructures ECFP enumerates and a fingerprint was built for.

RAN IT → NEXT CARD

That "cheap next try" (ECFP-counts + physchem + L1) and the deferred neural net were both run — see the next card. Short version: **N is the ceiling, not the encoder or the head.**

We gave structure its best shot — and finally ran the deferred neural net. Both confirm the setup.

Two "did we under-model this?" probes: a richer counts+physchem structure baseline, and the regularized multitask MLP the torch env had blocked. Neither overturns the result — and the MLP is decisively worse.

0.763

counts+physchem
structure (vs 0.757
ECFP)

0.725

raw + L1 sparse head
— worse than 0.763

0.680

multitask MLP fusion
(vs 0.766 logistic)

survives

GE benefit holds, still
SR-tilted

Why it all points one way. A richer representation lifts structure just +0.006 (inside the CI); L1 on 2066 raw features is *worse*, because N=177 can't estimate that many sparse coefficients; and the MLP — small and heavily regularized — still loses on every arm, overfitting generic variance. Tellingly the MLP's SR-specificity collapses (SR +0.060 $\approx$ NR +0.057): the linear head isn't just stronger here, it's the one that keeps the *mechanism legible*. **N is the ceiling, not the encoder or the head** — which points the lever back at data comparability and the rat→human layer.

More data didn't help — matched data helped, in proportion to how comparable it was.

Pooling a second rat dataset (TG-GATEs, +79 compounds) with a *mismatched* exposure window (hours) washed the signal out. The *time-matched* version (days) recovered it partway. The recovered signal tracks cross-dataset agreement.

N=177 · CLEAN

+.025

SR Δ AUC

single protocol

+256 · HOURS

0.38

SR $\rightarrow$ **+.001**

ComBat agreement r · washed out

+256 · DAYS

0.44

SR $\rightarrow$ **+.013**

ComBat agreement r · recovered

Why. Even same-platform, same-organ rat liver signatures for identical molecules agree only $r \approx 0.44$ after batch correction — a 24-hour liver and a 5-day liver are different biological states. **The bottleneck is comparability, not sample size.** It's the measurable, in-rat preview of the animal $\rightarrow$ human gap.

Val A

REFRAMED

Held-out Tox21 is infeasible by construction — the cross-validation already is that test.

Every rat compound with a Tox21 label is, by definition, in the supervised set (it's the intersection). There is no external hold-out — so the out-of-fold CV already answers it.

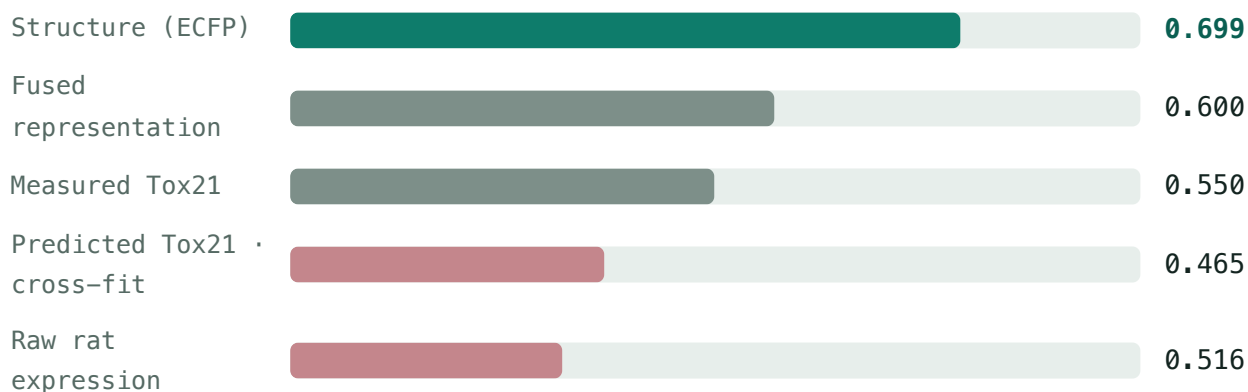
Why. The supervised set is $(\text{TG-GATEs} \cup \text{DrugMatrix}) \cap \text{Tox21}$. Honest to say so rather than manufacture a set that doesn't exist.

Val B

STRUCTURE WINS

For real human liver injury (DILIrank), the model adds nothing over structure.

A leakage-safe ladder on 134 shared drugs. The sanity gate passed first — measured Tox21 $\rightarrow$ DILI $\approx$ chance, reproducing the published near-random result — then structure beat everything.



The decisive check. The chained pipeline (predicted Tox21 $\rightarrow$ DILI) looked promising at 0.646 — but that was **leakage**. Regenerating the Tox21 predictions *out-of-fold* collapsed it to 0.465 $\approx$ chance. A ~ 0.12 -AUC illusion only cross-fitting exposes.

For market withdrawal, the pattern flips: biology beats structure.

A separate, noisier target (ChEMBL withdrawal status). Here structure is near-chance — withdrawal is driven by in-vivo/clinical effects, not molecular shape — and Tox21/expression features win.

0.56structure ($\approx$ chance)**0.65**

measured Tox21

Treat as a lead, not a headline. Only 35 withdrawn positives, a heterogeneous label (drugs are pulled for efficacy and commercial reasons too), and the predicted-Tox21 number swings between encoders — so it's a genuine, honest *contrast* worth chasing with a larger tox-specific set.

05

What it all says

yes · small

Does rat expression add to structure for Tox21?

A modest, stress-response-specific, encoder-robust benefit — but fragile: it needs feature balancing and a regularized head.

no

Does more animal data strengthen it?

No. Cross-dataset comparability caps out at $r \approx 0.44$; matched exposure recovers signal, mismatched destroys it.

no

Does it predict real human liver injury?

No. Structure alone wins; the one promising result was leakage, caught by cross-fitting.

Does it predict drug withdrawal?

Biology edges structure when the outcome is about in-vivo effect — but the label is noisy and the result unstable.

The animal→human gap is a *harmonisation* problem — not a data-volume one.

Two rat datasets for the same molecules agree only moderately; a human-cell-line assay panel doesn't predict clinical DILI; a rat transcriptome doesn't rescue it. The lever that matters is careful alignment of responses across systems — which is exactly what a rat→human translation layer is for, and what this pipeline is built to drop it into.

06

The next step — a rat→human translation layer

The project didn't just fail to close the gap — it *localised* it to one addressable step, and shipped the socket to plug the fix into.

Every experiment converged on one diagnosis: the ceiling is **comparability across biological systems, not data volume** — we measured it at $r \approx 0.44$ even for same-species, same-organ, same-molecule rat signatures. The animal→human gap is the same harmonisation problem, one system out. So the highest-leverage next move is a *learned* rat→human alignment, not more animal data.

TIER 1

Linear alignment

weeks · public data

Learn a linear rat→human map from an anchor set of compounds measured in *both* species (Open TG-GATEs human hepatocytes, LINCS L1000), over 1:1 orthologs. CORAL / optimal transport.

TIER 2

Contrastive encoder

a GPU-month

Project rat and human signatures into a shared latent where the same compound's responses are neighbours and a species discriminator can't separate them. Where $r \approx 0.44$ could actually be pushed.

TIER 3

Pathway translation

the ambitious version

Align at Reactome / MSigDB pathway activity, not raw genes — platform-robust, interpretable, and tied to the SR hypothesis: conserved stress pathways translate; species-divergent metabolism may not.

THE FALSIFIABLE PREDICTION THAT MAKES IT FUNDABLE

If the animal→human gap is truly **harmonisation**, a translation layer should lift the human-DILI **expression** arm from its current chance-level **0.52** toward structure's **0.70** — and help SR mechanisms more than NR, mirroring the in-rat result. If it *doesn't* move 0.52, the gap is deeper than harmonisation — itself a publishable negative.

The embed () interface is already swappable (a RatToHumanEmbedder stub with the identical contract) — the model drops in with zero change to the CV harness or fusion. "Funded" buys anchor-set curation and encoder compute, not another animal study.

07

Honest limitations

Stated plainly, because the checks that surface them are the point.

Small N is the binding constraint.

177 compounds (256 pooled); sparse assays like NR-PPAR γ have 12 actives. Richer features, an L1 head, GBM and a multitask MLP all failed to beat the regularized linear model at this N.

Baseline below SOTA.

ECFP+logistic (stronger than frozen ChemBERT and counts+physchem here) still sits under published DILI models (0.75–0.83); a *fine-tuned* transformer is the open follow-up.

Tox21 is a prior, not truth.

Human in-vitro assays are a mechanism prior — we predict them, not the rat's own histopathology.

Imbalanced, noisy clinical labels.

DILIrank overlap is ~82% positive; the withdrawal label mixes reasons and has only 35 positives.

REPRODUCIBLE

Every result regenerates from the scripts plus public downloads — all transforms fit in-fold, seeds fixed, splits persisted. Large matrices and third-party raw labels are not redistributed; the download URLs live in the scripts.

[GitHub](#)[HuggingFace](#)

Cross-Species Translational Alignment

TG-GATEs · DrugMatrix · Tox21 · DILIrank · ChEMBL

Building an AI Scientist — London, Jul 2026

Data: NTP · LSDB · EPA/NCATS · FDA/NCTR · EBI